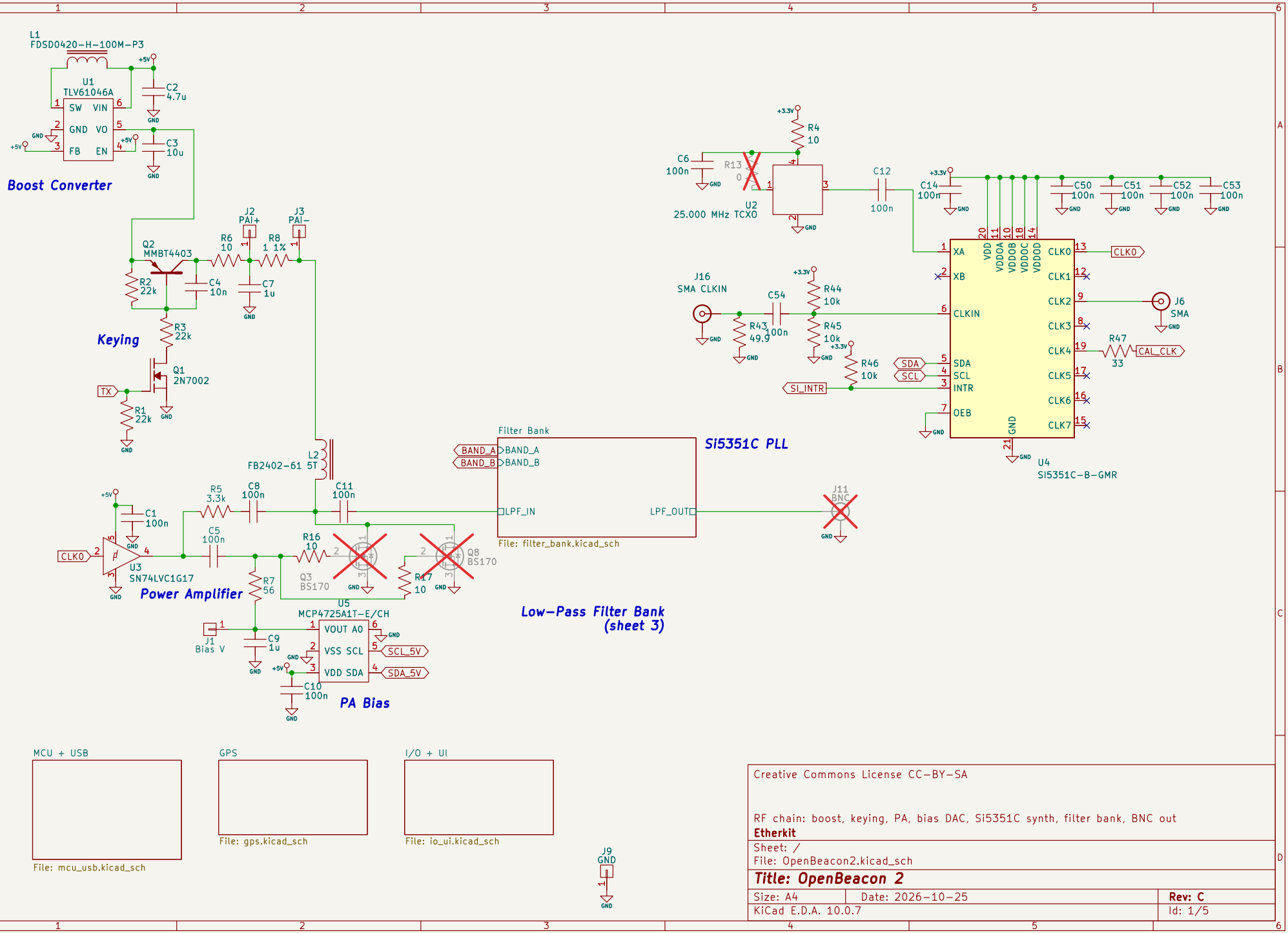
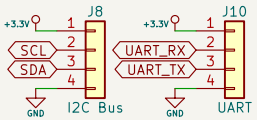
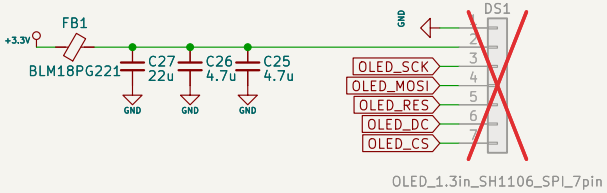

L1
FDSD0420-H-100M-P3
U1
TLV61046A
SW VIN
GND VO
FB EN
C2
4.7u
C3
10u
Boost Converter
Q2
MMBT4403
R2
22k
R3
22k
C4
10n
R6
10
R8
1 1%
C7
1u
J2
PAI+
J3
PAI-
Keying
Q1
2N7002
TX
R1
22k
L2
FB2402-61 5T
R5
3.3k
C8
100n
C11
100n
C1
100n
C5
100n
CLK0
U3
SN74LVC1G17
Power Amplifier
R16
10
R7
56
Q3
BS170
Q8
BS170
R17
10
U5
MCP4725A1T-E/CH
VOUT A0
VSS SCL
VDD SDA
SCL_5V
SDA_5V
J1
Bias V
C9
1u
C10
100n
PA Bias
Filter Bank
BAND_A
BAND_B
LPF_IN
LPF_OUT
File: filter_bank.kicad_sch
Low-Pass Filter Bank
(sheet 3)
J11
BNC
R4
10
C6
100n
R13
0
U2
25.000 MHz TCXO
C12
100n
C14
100n
C50
100n
C51
100n
C52
100n
C53
100n
J16
SMA CLKIN
C54
100n
R43
49.9
R44
10k
R45
10k
R46
10k
SDA
SCL
SI_INTR
XA
XB
CLKIN
OEB
VDD
VDDA
VDDOB
VDDOC
VDDOD
CLK0
CLK1
CLK2
CLK3
CLK4
CLK5
CLK6
CLK7
J6
SMA
R47
33
CAL_CLK
U4
SI5351C-B-GMR
Si5351C PLL
MCU + USB
File: mcu_usb.kicad_sch
GPS
File: gps.kicad_sch
I/O + UI
File: io_ui.kicad_sch
J9
GND
Creative Commons License CC-BY-SA
RF chain: boost, keying, PA, bias DAC, Si5351C synth, filter bank, BNC out
Etherkit
Sheet: /
File: OpenBeacon2.kicad_sch
Title: OpenBeacon 2
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 1/5

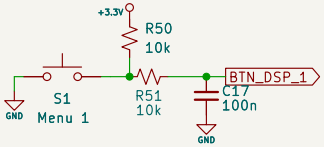


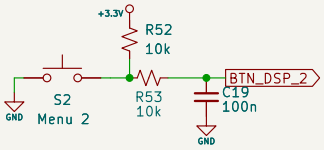
IO Headers

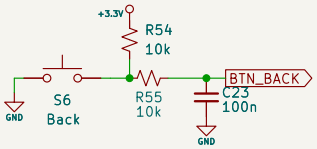


128x64 OLED (1.3in SH1106, SPI)

Same module and pinout as Si5351AVFO DS1: DNP, kit-supplied, user-soldered. 3.3 V logic, no level shift.
FB1 + C25–C27 copy the VFO OLED rail filter at the header.

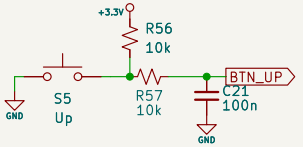


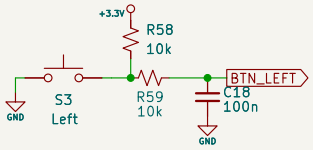


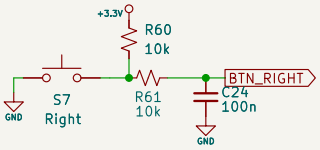


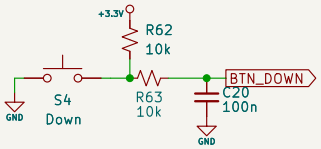
Function keys (under display)

Layout: S1 / S2 / S6 in one row directly under DS1 (as on Si5351AVFO). D-pad cross: S5 Up, S4 Down, S3 Left, S7 Right. Each line: 10k pull-up, 10k series, 100n at the MCU pin (Si5351AVFO network).

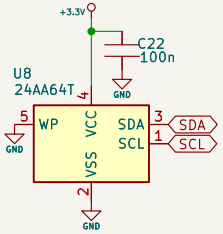




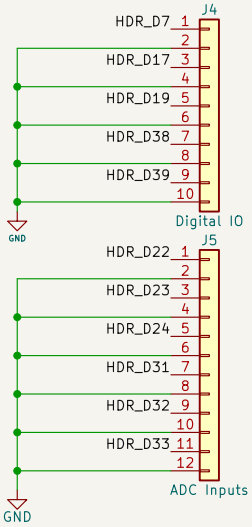


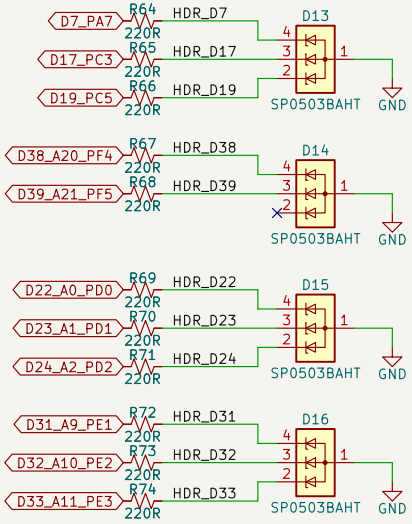


D-pad



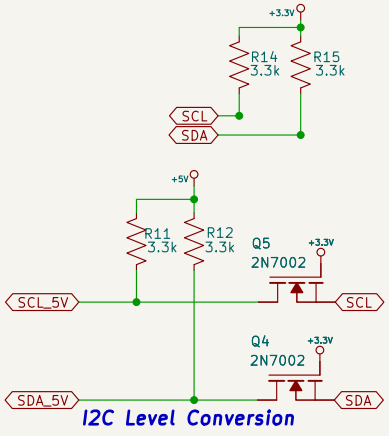
External EEPROM





GPIO Headers (DxCore pin names)

ESD: TVS at the header pin, 220R in series to the MCU (limits clamp current into the AVR pin, R x 30 pF = 7 ns)



I2C Level Conversion

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I/O headers, SH1106 SPI OLED, 7 buttons, I2C level shift, 24AA64

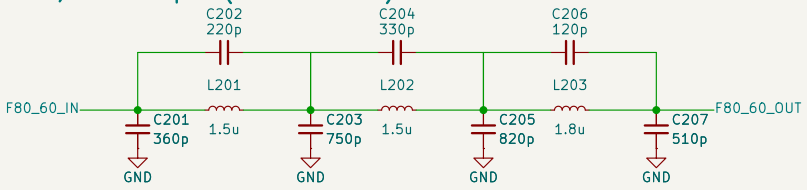
Etherkit

Sheet: /I/O + UI/
File: io_ui.kicad_sch

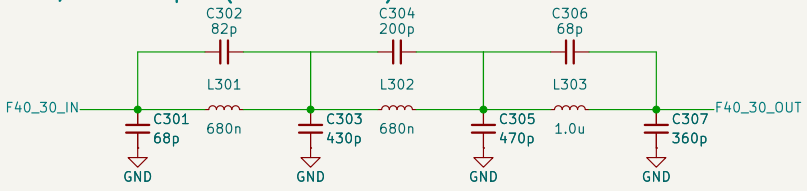
Title: OpenBeacon 2 – I/O and User Interface

Size: A4	Date: 2026-10-25	**Rev: C**
KiCad E.D.A. 10.0.7		Id: 2/5

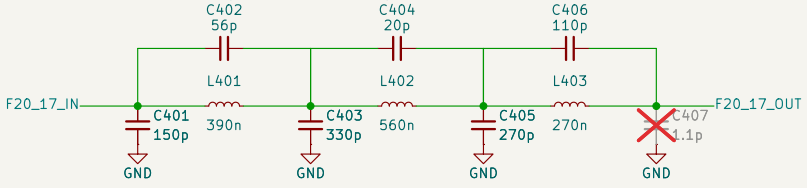
F2 80/60 m low-pass (7th-order Cauer)



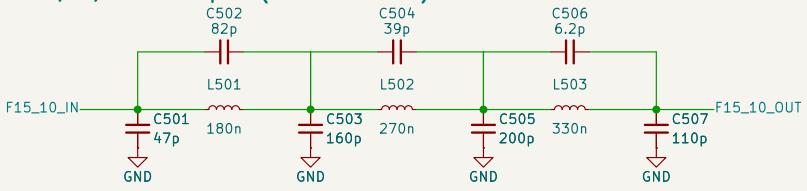
F3 40/30 m low-pass (7th-order Cauer)



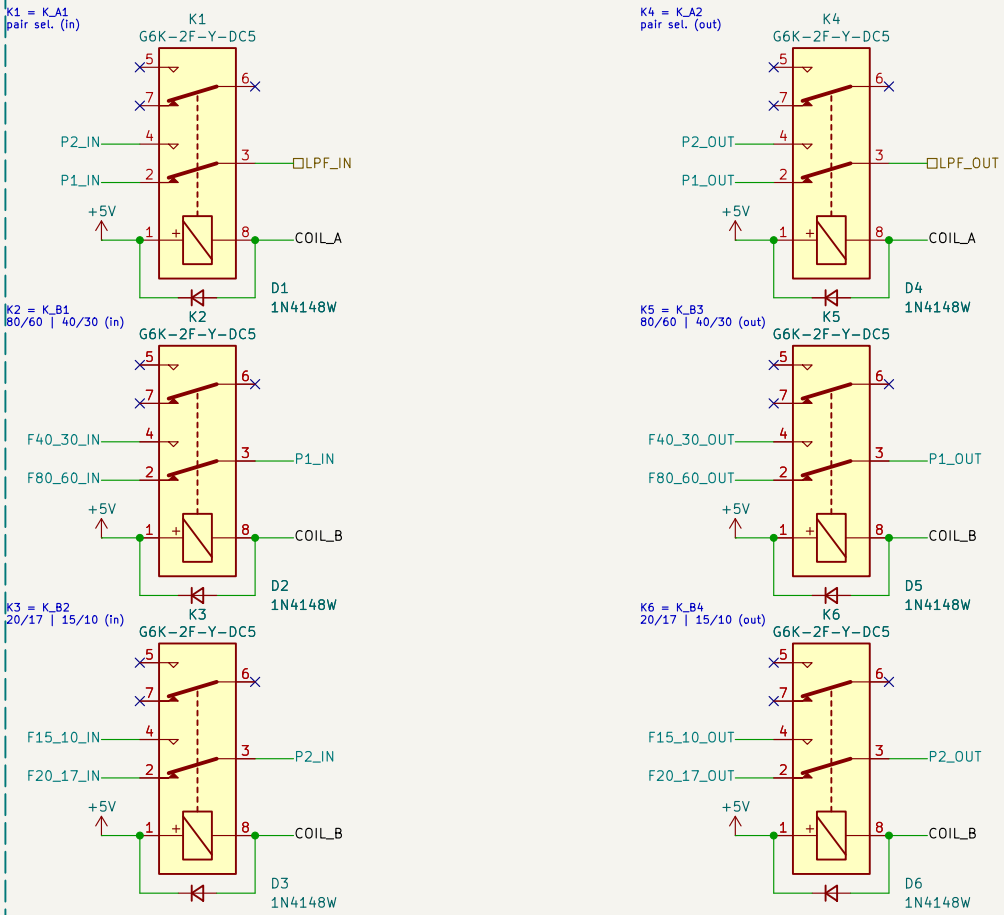
F4 20/17 m low-pass (7th-order Cauer)



F5 15/12/10 m low-pass (7th-order Cauer)



Relay tree (2 levels, input + output mirrored; all off = 80/60 m)



Relay drivers



BAND_A -> K1/K4, BAND_B -> K2/K3/K5/K6.
Q6 (BAND_A): 2 coils, 42 mA nom (5 V / 237 R), <= 49 mA at 5.25 V, -10 % coil R.
Q7 (BAND_B): 4 coils, 84 mA nom, <= 98 mA.
AO3400A: 5.7 A cont., 48 mR @ Vgs 2.5 V -> <5 mV drop.
One 1N4148W flyback per coil, at each relay.

Band code + notes

Band code (BAND_A, BAND_B):
0 0 80/60 m (all relays off, NC path, default)
0 1 40/30 m (K2, K3, K5, K6 on)
1 0 20/17 m (K1, K4 on)
1 1 15/12/10 m (all six on)
Unselected filters float (open at both ends).
+5V relay load: 0 / 84 / 42 / 127 mA nominal (<=147 mA).
Relay pins: COM right, NC lower left, NO upper left; pole 2 unused.
Input and output paths never share a relay body.
Layout: keep K1-K2-K3 and K4-K5-K6 traces short, 50 ohm CPWG over solid GND, filters next to their relays.

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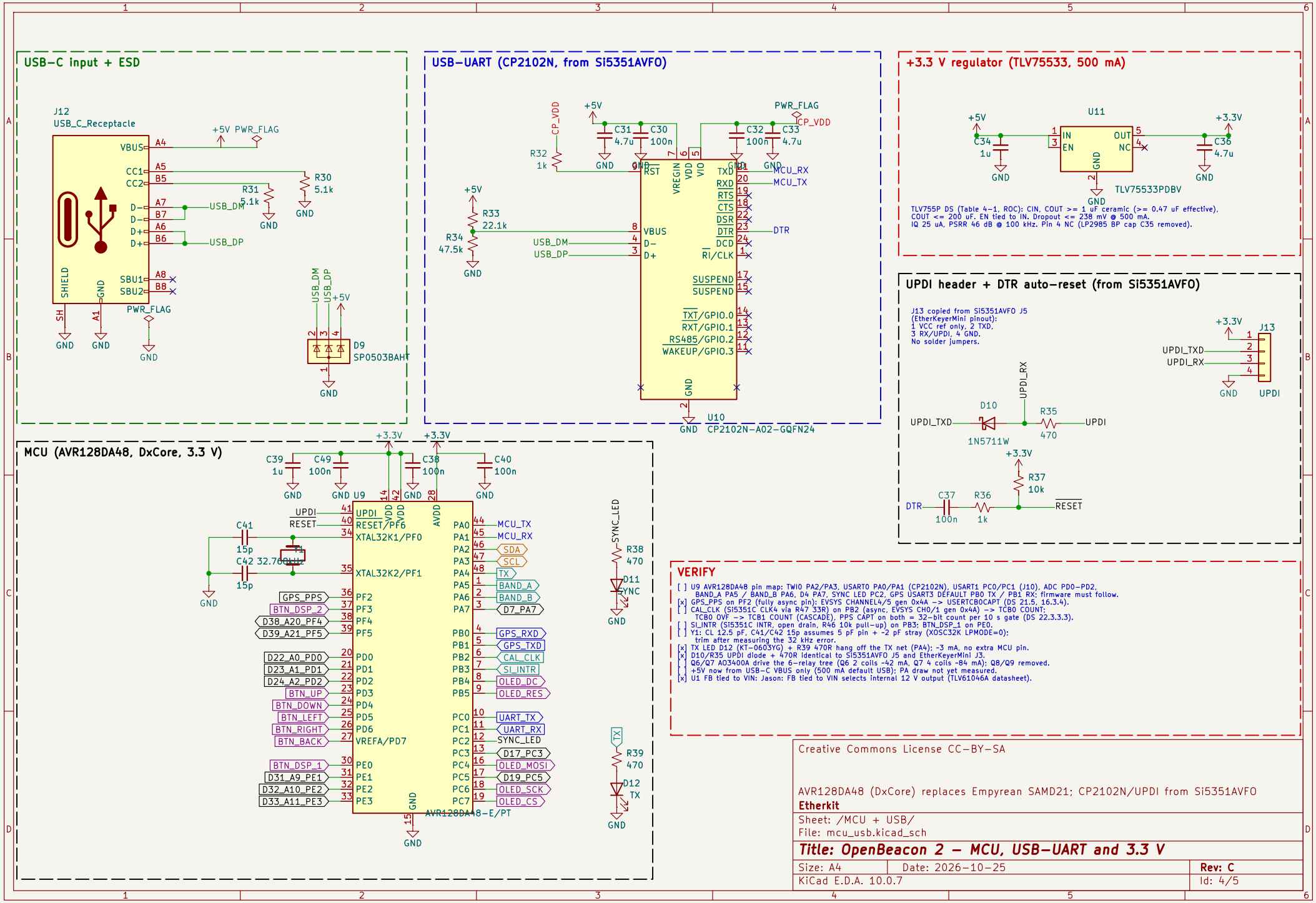
4 7th-order Cauer LPFs, 6-relay 2-level tree (2-bit BAND_A/BAND_B); unselected filters open

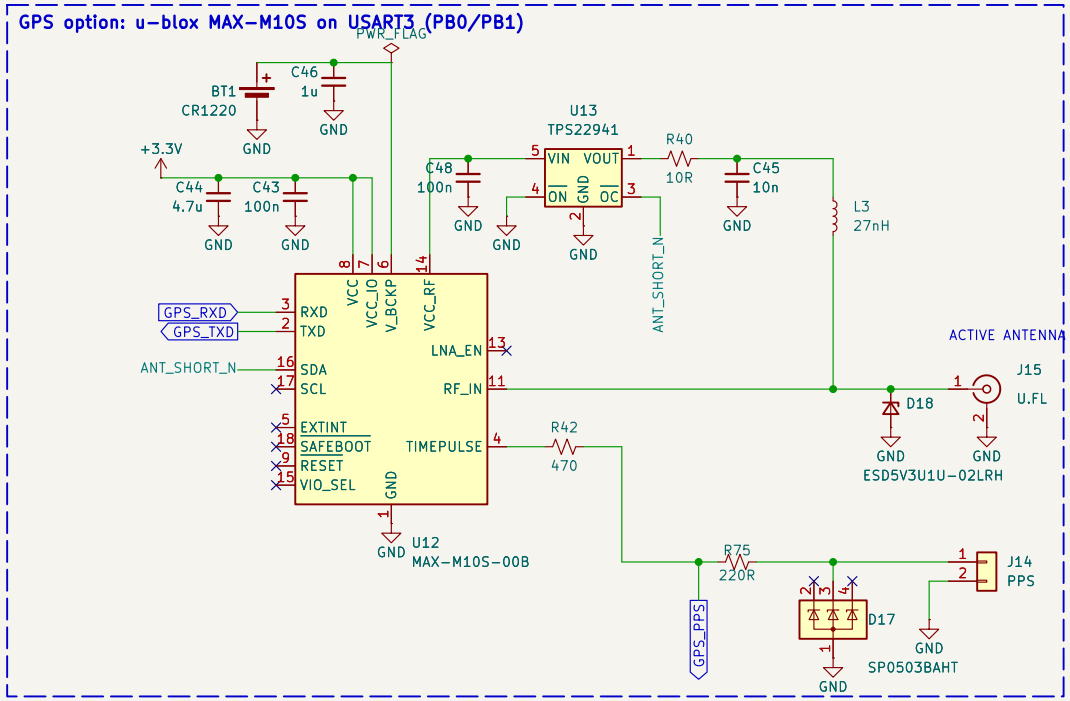
Etherkit

Sheet: /Filter Bank/
File: filter_bank.kicad_sch

Title: OpenBeacon 2 – Low-Pass Filter Bank

Size: A4	Date: 2026-10-25	Rev: C
KiCad E.D.A. 10.0.7		Id: 3/5


USB-C input + ESD
J12
USB_C_Receptacle
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
SH
A1
PWR_FLAG
+5V
R30
5.1k
R31
5.1k
USB_DM
USB_DP
D9
SP0503BAHT
USB-UART (CP2102N, from Si5351AVFO)
CP_VDD
R32
1k
C31
4.7u
C30
100n
C32
100n
C33
4.7u
R33
22.1k
R34
47.5k
VBUS
D-
D+
RST
VREGIN
VDD
VIO
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
WAKEUP/GPIO.3
MCU_RX
MCU_TX
DTR
U10
CP2102N-A02-GQFN24
+3.3 V regulator (TLV75533, 500 mA)
U11
C34
1u
C36
4.7u
IN
EN
OUT
NC
TLV75533PDBV
TLV755P DS (Table 4-1, ROC): CIN, COUT >= 1 uF ceramic (>= 0.47 uF effective),
COUT <= 200 uF. EN tied to IN. Dropout <= 238 mV @ 500 mA.
IQ 25 uA, PSRR 46 dB @ 100 kHz. Pin 4 NC (LP2985 BP cap C35 removed).
UPDI header + DTR auto-reset (from Si5351AVFO)
J13 copied from Si5351AVFO J5
(EtherKeyerMini pinout):
1 VCC ref only, 2 TXD,
3 RX/UPDI, 4 GND.
No solder jumpers.
J13
UPDI_TXD
UPDI_RX
UPDI
D10
1N5711W
R35
470
R37
10k
C37
100n
R36
1k
DTR
RESET
MCU (AVR128DA48, DxCore, 3.3 V)
C39
1u
C49
100n
C38
100n
C40
100n
C41
15p
C42
15p
32.768kHz
U9
AVR128DA48-E/PT
UPDI
RESET/PF6
XTAL32K1/PF0
XTAL32K2/PF1
PF2
PF3
PF4
PF5
PD0
PD1
PD2
PD3
PD4
PD5
PD6
VREFA/PD7
PE0
PE1
PE2
PE3
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB3
PB4
PB5
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
MCU_TX
MCU_RX
SDA
SCL
TX
BAND_A
BAND_B
D7_PA7
GPS_RXD
GPS_TXD
CAL_CLK
SI_INTR
OLED_DC
OLED_RES
UART_TX
UART_RX
SYNC_LED
D17_PC3
OLED_MOSI
D19_PC5
OLED_SCK
OLED_CS
GPS_PPS
BTN_DSP_2
D38_A20_PF4
D39_A21_PF5
D22_A0_PD0
D23_A1_PD1
D24_A2_PD2
BTN_UP
BTN_DOWN
BTN_LEFT
BTN_RIGHT
BTN_BACK
BTN_DSP_1
D31_A9_PE1
D32_A10_PE2
D33_A11_PE3
R38
470
D11
SYNC
R39
470
D12
TX
VERIFY
[ ] U9 AVR128DA48 pin map: TWI0 PA2/PA3, USART0 PA0/PA1 (CP2102N), USART1 PC0/PC1 (J10), ADC PD0-PD2,
BAND_A PA5 / BAND_B PA6, D4 PA7, SYNC LED PC2, GPS USART3 DEFAULT PB0 TX / PB1 RX; firmware must follow.
[x] GPS_PPS on PF2 (fully async pin): EVSYS CHANNEL4/5 gen 0x4A -> USERTCBOCAPT (DS 21.5, 16.3.4).
[ ] CAL_CLK (SI5351C CLK4 via R47 33R) on PB2 (async, EVSYS CH0/1 gen 0x4A) -> TCB0 COUNT;
TCB0 OVF -> TCB1 COUNT (CASCADE), PPS CAPT on both = 32-bit count per 10 s gate (DS 22.3.3.3).
[ ] SI_INTR (SI5351C INTR, open drain, R46 10k pull-up) on PB3; BTN_DSP_1 on PE0.
[ ] Y1: CL 12.5 pF, C41/C42 15p assumes 5 pF pin + ~2 pF stray (XOSC32K LPMODE=0);
trim after measuring the 32 kHz error.
[x] TX LED D12 (KT-0603YG) + R39 470R hang off the TX net (PA4); ~3 mA, no extra MCU pin.
[x] D10/R35 UPDI diode + 470R identical to Si5351AVFO J5 and EtherKeyerMini J3.
[ ] Q6/Q7 AO3400A drive the 6-relay tree (Q6 2 coils ~42 mA, Q7 4 coils ~84 mA); Q8/Q9 removed.
[ ] +5V now from USB-C VBUS only (500 mA default USB); PA draw not yet measured.
[x] U1 FB tied to VIN: Jason: FB tied to VIN selects internal 12 V output (TLV61046A datasheet).
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AVR128DA48 (DxCore) replaces Empyrean SAMD21; CP2102N/UPDI from Si5351AVFO
Etherkit
Sheet: /MCU + USB/
File: mcu_usb.kicad_sch
Title: OpenBeacon 2 - MCU, USB-UART and 3.3 V
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 4/5



VERIFY

- [x] Antenna short: TPS22941 limit 40–80 mA, auto-restart, avg <= 27 mA (was 320 mA).
- [x] V_BCKP: CR1220 in KH-BS1220-2-SMT (C5365932): C47/D13/R41 removed.
- [ ] BT1 footprint: draw from KH-BS1220-2-SMT DS.
- [ ] U13 TPS22941 JLC stock thin (26 pcs).
- [ ] Short peak 80 mA > VCC_RF 50 mA op. (DS T13) for <= 20 ms bursts; confirm on bench.
- [ ] V_IO ramp 25–35000 us/V (DS 4.1): measure.
- [ ] GPS_PPS MCU pin: see mcu_usb.

RECEIVER: VCC + V_IO tied to +3.3V; VIO_SEL open = 3.3 V I/O (IM 4.1.2).
C43/C44 beside pins 7/8, short GND vias; no series R > 0.2 ohm in VCC (IM 4.1.1).
RESET_N, EXTINT, SCL, SAFEBOOT_N, LNA_EN left open (DS Table 10).
SDA = ANT_SHORT_N (IM 3.3.1.2): CFG-HW-ANT_CFG_SHORTDET=1, ANT_SUP_SHORT_PIN=SDA PIO.
PPS: TIMEPULSE = SAFEBOOT_N via 1k inside (DS fn 15); nothing may hold it low at power-up, so no pull-down. R42 limits a shorted J14 to ~7 mA (I_PIO 10 mA).

ANTENNA: RF_IN is DC-blocked inside the module (IM Table 1), no external DC block.
Bias-T VCC_RF -> U13 TPS22941 -> R40 10R -> C45 10n -> L3 27 nH onto RF line (IM 3.3.1.2, 4.3.4).
U13: I_LIM 40–80 mA, 5–20 ms blanking, 40–160 ms restart (TPS2294x DS 7.5): short avg <= 27 mA, typ 7 mA (VCC_RF abs max 250 mA, DS Table 12). 25 mA antenna drop ~0.27 V -> ~2.9 V at J15.
Layout: 50 ohm CPWG pin 11 -> J15, L3 pad on the line, >= 5 mm from digital.
BACKUP: BT1 CR1220 (3 V, 35 mAh, user-fitted, NOT rechargeable, no charger) on V_BCKP;
C46 1u at pin 6 (IM 4.1.3). 35 mAh / 28 uA (DS Table 17) = ~1250 h (~52 days) powered off.

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MAX-M10S on USART3 (PB0/PB1), PPS to J14, current-limited active-antenna bias-T, CR1220 V_BCK

Etherkit

Sheet: /GPS/
File: gps.kicad_sch

Title: OpenBeacon 2 – GPS (u-blox MAX-M10S)

Size: A4	Date: 2026-10-25	**Rev: C**
KiCad E.D.A. 10.0.7		Id: 5/5